

# Qingqiao Cai

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## EDUCATION

Sun Yat-sen University | Shenzhen, China

09/2022 - 07/2027 (Expected)

- B.Eng. in **Biomedical Engineering**, GPA: 3.7/4.0.
- **Scholarship**: Third-Class Scholarship for Outstanding Students awarded 3 times (2022-2025).
- **Skills**:
  - **Software & Engineering Tools**: KiCad, SolidWorks, AutoCAD, MATLAB, Proteus, Keil uVision, Arduino IDE, VS Code, Mimics Research, Geomagic Wrap.
  - **Hardware & Embedded Systems**: Schematic capture, 2-/4-layer PCB layout, analog front-end design, component selection, power management, hardware debugging; ESP32-S3, Arduino Nano, AT89C52.
  - **Programming & Data**: C/C++, Python, MATLAB; logistic regression, SVM, bagged trees, and CNN model development/training.
  - **Electrochemistry & Sensing**: Microneedle fabrication, enzyme-electrode fabrication, electrochemical deposition, chronoamperometry, and cyclic-voltammetry characterization.

## RESEARCH EXPERIENCE

**Wearable Microneedle Bioelectronic System for Continuous Health Monitoring** | Notre Dame, United States

*Independent Researcher* | Advisor: Prof. Kaiyu Fu, Department of Chemistry & Biochemistry, University of Notre Dame

07/2026 - Present

- Defined a 2-module wearable architecture including microneedle sensing front-end and an arm-cuff unit integrating the MCU, battery management, wireless charging, and Bluetooth communication
- Designed the comprehensive system schematics and a 4-layer PCB for 6 sensing channels, including 2 electrochemical channels with microneedle interfaces and 4 open-circuit potential channels, and a 2-layer PCB for compact power management
- Fabricated polymer microneedle arrays through high-resolution 3D-printed mold design, mold replication, and polymer casting, establishing the minimally invasive sensor-tissue interface for continuous biochemical sensing
- Developing desktop and mobile applications for real-time data acquisition, visualization, wireless communication, data logging, and device control

**Wearable EEG-EOG Monitoring System for Human-Computer Interaction** | Shenzhen, China

*Independent Researcher* | Advisor: Prof. Jiyu Li, School of Biomedical Engineering, Sun Yat-sen University

07/2025 - Present

- Independently designed and iterated a 4-layer ADS1299- and ESP32-S3-based acquisition PCB for an 8-channel wearable electrophysiological platform, supporting simultaneous 6-channel EEG and 2-channel EOG recording for vertical and horizontal eye movements, with reserved interfaces for flexible patch electrodes and portable system integration
- Built and validated 3 core hardware subsystems, including the ADS1299 analog front end, low-noise power-supply circuitry, and ESP32-S3 wireless communication module, establishing a complete signal path from 8 electrode inputs to signal acquisition, digitization, and wireless transmission
- Implemented a real-time BLE-based data transmission workflow for on-device packetization, wireless streaming, and host-side reception of electrophysiological signals
- Developing a MATLAB analysis pipeline for signal preprocessing, feature extraction, and model development, supporting quantitative analysis of EEG/EOG recordings and subsequent eye-movement and related electrophysiological applications

**Implantable BCI for Large-Scale Neural Recording** | Shenzhen, China

*Member* | Advisor: Prof. Xudong Lin, School of Biomedical Engineering, Sun Yat-sen University

03/2025 - 05/2025

- Designed PDMS-based structures for an implantable brain-computer interface for large-scale neural recording, including guide channels and magnetic head components
- Improved PDMS perforation by identifying needle insertion prior to PDMS curing as an effective process to obtain smooth-walled and dimensionally consistent channels
- Implemented template-based needle fixation methods to enable batch fabrication of multiple guide channels with improved reproducibility
- Fabricated tent-shaped electrode scaffolds using micro/nano stereolithography 3D printing (Mofang S140) to assess structural feasibility for large-area neural applications

**Neural Conduit Fabrication and Structural Control** | Shenzhen, China

*Member* | Advisor: Prof. Chao Zhang, School of Biomedical Engineering, Sun Yat-sen University

08/2024 - 09/2024

- Established control over pore structure and permeability in neural conduit walls through systematic variation of PEG molecular weight, solution concentration, and dip-coating parameters
- Characterized the dissolution behavior of PEG aqueous solutions (1k-20k; 1-2%) and established thickness calibration by designing a withdrawal speed and coating-cycle parameter matrix

- Evaluated pore morphology and early-stage degradation through multi-field microscopic imaging, pore-size statistics, and dry-weight measurements
- Constructed a variable-speed withdrawal setup using microfluidic equipment to address coating nonuniformity on a single mandrel and improve the consistency of multi-cycle coatings

**PUBLICATION**

Sun, J.#, Liu, M.#, Jiang, C.#, **Cai, Q.**, Cai, R., Li, J., Xiao, Y., Xian, Q., Sun, X., Jiang, L., Li, C., Tso, C. Y., Yu, X., Zhou, Y., Peng, Z., & Li, J. Self-stratified Stretchable Passive Cooling Interface for Thermal Management of On-Skin Electronics. *Soft Science*, 6, 43 (2026). [DOI: 10.20517/ss.2026.33](https://doi.org/10.20517/ss.2026.33)

**ACADEMIC PROJECTS**

**Electrochemical Sweat Glucose Sensing System**

Course Project

10/2025 - 12/2025

- Built and evaluated an enzyme-based electrochemical sensing system for sweat glucose detection, integrating Prussian, Blue-modified enzyme electrodes, signal-conditioning hardware, Arduino Nano acquisition, Bluetooth transmission, and an Android app for real-time monitoring
- Led development of the signal-conditioning hardware by designing a configurable transimpedance front end with 9 gain settings using a  $3 \times 3$  switch configuration, enabling amplification and current-to-voltage conversion across different electrochemical current ranges
- Developed an Android app for real-time Bluetooth reception, visualization, and data logging, enabling simultaneous monitoring of sensor output and glucose concentration during electrochemical testing
- Fabricated Prussian Blue-modified electrodes through electrochemical deposition and immobilized glucose oxidase, followed by standardized electrode storage, cyclic-voltammetry characterization, and performance testing
- Performed chronoamperometric measurements at 0.1 V and calibrated the sensor at 4 glucose concentrations of 1.0, 2.5, 5.0, and 7.5 mM, using approximately 30 s of mixing and stabilization after each addition before steady-state readout

**Feature-Based Classification of Neurocritical Care Patients Using TCD Signals**

Course Project

10/2025 - 12/2025

- Processed 31 TCD recordings from 18 subjects, including 6 healthy participants and 12 neurocritical-care patients, using STFT-based spectrogram construction, spectral enhancement, adaptive envelope extraction, beat segmentation, and SQI filtering to obtain reliable beat-level representations from raw IQ signals
- Extracted features across 3 domains, including time-domain, PSD, and entropy-based characteristics, and evaluated individual and fused feature sets using Jarque-Bera testing, Mann-Whitney U testing, and FDR correction to identify statistically significant differences between healthy and patient groups
- Built and compared 3 traditional machine-learning classifiers, including logistic regression, RBF SVM, and bagged trees, using stratified 5-fold cross-validation; the best PSD-based logistic regression model achieved a ROC-AUC of  $0.978 \pm 0.050$
- Designed a beat-based 1D CNN using 256-point inputs and 4 convolutional blocks with 16, 32, 64, and 128 filters, and applied recording-wise 5-fold cross-validation with recording-level prediction aggregation to evaluate model performance while preventing data leakage across recordings
- Implemented the complete analysis workflow in MATLAB, integrating raw-signal preprocessing, beat segmentation and quality control, feature engineering, statistical analysis, traditional machine learning, CNN training, performance visualization, and automated result generation

**CAMPUS LEADERSHIP**

**Swimming Association of Sun Yat-sen University | Shenzhen, China**

Department Head

09/2023 - 07/2024

- Led a 10-member team to run weekly swimming training sessions for university students, focusing on technique, safety, and skill progression
- Organized member recruitment and monthly workshops to train and assess instructors in teaching methods and class management
- Supported planning and promotion of the university-wide "Kangle Cup" swimming competition, which attracted more than 100 participants

**COMPETITIONS & HONORS**

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|-------------------------------------------------------------------------------------------------------------|----------------|
| <b>National Competition Bronze Award</b> , 2025 China International College Students Innovation Competition | <b>11/2025</b> |
| <b>Provincial-Level Gold Award</b> , 2025 China International College Students Innovation Competition       | <b>08/2025</b> |
| <b>Team Champion</b> , 2024 Four-University Baseball and Softball Tournament, Sun Yat-sen University        | <b>05/2024</b> |

**ADDITIONAL INFORMATION**

**Hobbies:** Badminton, Chinese Classical Dance (Level 10)